



CPT316: Programming Language Implementation and Paradigms
Semester 1 2025/2026
Assignment 2

Weightage: 15%

Presentation: Week 12 (21/12/2025-27/12/2025) and Week 13 (28/12/2025-3/1/2026)

Due Date for Assignment Submission : (3/1/2026 time before 5.00PM)

Submission: Submit via the elearn@USM by the due date.

Assignment Type: Group Work (4 members per group)

Assignment Overview

The goal of this assignment is to design and implement a system utilizing a programming language and framework. Students will use two (2) programming language or paradigm approaches in this assignment. Students are required to develop an application using JavaScript on Node-RED (compulsory). This application must then be connectable and able to communicate with another platform that uses a different programming language, such as Python or C++.

3.Submission Guidelines

Submit the entire project as a single zip file. The zip file should contain:

1. The source code of software.
2. The report is in PDF format.
3. Presentation

Note:

Projects are not constrained in any way, and the importance of a project is that it can demonstrate the capacity to use two paradigms. The project may be a full or partial package or module for project development.

CLO 3: Differentiate various programming language designs and implementations in order to select the best programming language in problem-solving process. (5%)															
	CLO	Criteria	Excellent (4)	Good (3)	Satisfactory (2)	Needs Improvement (1)	Weight (w)	Score (s)				Marks (s x w)			
								1	2	3	4	1	2	3	4
1	CLO3-PL03	Understanding of Concepts	Demonstrates a deep understanding of key concepts with clear, accurate explanations.	Demonstrates a good understanding of concepts with few errors or omissions.	Shows a basic understanding but lacks clarity or depth in some areas.	Demonstrates little or no understanding of the concepts; significant errors.	3								
2	CLO3-PL03	Application of Knowledge	Applies concepts accurately to real-world situations or problem-solving.	Applies concepts adequately but may lack creativity or depth.	Attempts to apply concepts but with errors or limited relevance.	Fails to apply concepts or demonstrates a misunderstanding of application.	3								
3	CLO3-PL03	Analysis and Critical Thinking	Provides detailed and well-organized analysis with evidence of critical thinking.	Shows good analytical thinking with some evidence of logical reasoning.	Analysis is limited and lacks depth; critical thinking is minimal.	Little to no analysis; fails to demonstrate critical thinking or reasoning.	3								
4	CLO3-PL03	Synthesis and Creativity	Combines ideas effectively to create innovative or original solutions.	Combines ideas logically but lacks originality or creativity.	Attempts synthesis, but ideas are disorganized or lack logical connections.	Fails to combine ideas coherently; no evidence of creativity.	2								
5	CLO3-PL03	Evaluation and Argumentation	Presents well-supported arguments or evaluations with relevant evidence.	Presents arguments with some evidence, but lacks clarity or persuasiveness.	Provides weak arguments or evaluations with minimal evidence.	Arguments are unsupported, unclear, or absent; little evidence is provided.	2								
6	CLO3-PL03	Organization and Clarity	Content is well-organized, clear, and flows logically with no redundancies.	Content is organized and mostly clear, but minor lapses in flow or clarity exist.	Content is somewhat disorganized, unclear, or repetitive.	Content is poorly organized, unclear, and difficult to follow.	2								
								Total							
								Total (5%)							

CLO4: Demonstrate the ability to find relevant information for a solid problem solution before development (10%)															
	CO-PO	Criteria	Poor	Fair	Good	Very Good	Weight (w)	Score (s)				Marks (s x w)			
			1.0 - 2.0	3.0 - 4.0	5.0 - 7.0	8.0 - 10.0		S1	S2	S3	S4	S1	S2	S3	S4
1	CO4-PL07	Task Delegation and Quality of Work	No clear task delegation in the group. Task delegation is not listed in the presentation. Work does not meet expectations.	Task delegation is listed but no teamwork and collaboration in the group. Work meets some expectations but with issues.	Task delegation is listed but it seems only certain member control the presentation. Work meets expectations.	Task delegation is listed, clear teamwork and the group always trying to answer questions eagerly. Work exceeds expectations.	3								
2	CO4-PL07	Presentation and Visual Aides	The presentation did not cover all important points in the design, and did not demonstrate any functionality of the project OR Visual aides are not provided.	The presentation failed to sufficiently cover all important points in the design, and did not demonstrate the full functionality of the project OR Visual aides are used, but lack the level of detail necessary and are not used to present points effectively.	The presentation sufficiently covers all important points in the design, as well as functionality of the project OR Visual aides are of high quality and sufficiently used to articulate points.	The presentation details all important points in the design , as well as functionality of the project OR Visual aides are interesting, of very high quality and used effectively to articulate points.	3								
3	CO4-PL07	Time Management	Presentation is completed significantly above the allocated time frame. Significant delays that take up most of the presentation time.	Presentation is completed moderately above the allocated time frame. Noticeable delays that significantly impede the overall efficiency of the time given.	Presentation is completed slightly above or below the allocated time frame. Noticeable delays, but does not significantly impede the overall efficiency of the time given.	Presentation is completed in the allocated time frame. Presentation time is utilized efficiently for presentation.	4								
								Total							
								Total (10%)							

CLO3 (5%)									
CLO4 (10%)									
Total (15%)									
Verified by									